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Process-Interruption Recovery and Performance Trade-offs in Three Game Save-File Update Strategies: A Reproducible Case Study

Zain Kassem

Independent researcher, Dubai, United Arab Emirates

* Corresponding author: zainkassem7@gmail.com

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Abstract- Local game saves can be left incomplete when a game process stops during an update. We compare direct overwrite, same-directory temporary-file replacement, and two-slot generation with SHA-256 validation using synthetic game-state payloads. A .NET harness and a separate Unity 6.3 Windows Player were interrupted at predeclared checkpoints on one Windows 11 workstation. Across three payload sizes, the .NET run recorded 3,500 fault observations and 900 timings; the Unity replication recorded 2,940 fault trials and 900 timings. All 20 trials in each checkpoint group agreed: overwrite had no valid save after truncation until the new payload was complete, while replacement and two-slot retained the old state until commit. No valid trial produced an unexpected state. Median save times ranged from 0.120–0.236 ms, 1.639–5.621 ms, and 0.251–0.632 ms across the three .NET procedures; Unity showed the same recovery pattern, with platform-specific costs. This bounded, reproducible comparison concerns application-process termination only and establishes neither power-loss durability nor cross-platform guarantees.

Keywords- Game save systems; crash recovery; file update strategy; fault injection; reproducible benchmarking

1. Introduction

Games routinely persist player progress, inventories, quests, and world state. An interruption during an update can leave a save unavailable or incomplete, while additional safeguards can increase save latency and storage use. Developers therefore face a practical choice among simple overwrite, temporary-file replacement, and redundant versioned saves. The choice is especially relevant for games whose saves are written locally and whose failure handling is implemented at the application layer.

Prior systems research shows that application-level recovery depends on the semantics of file operations and the persistence behavior of the underlying file system. Pillai et al. demonstrated that crash-consistency protocols can rely on subtle file-system properties and that these properties vary across systems [1]. Bornholt et al. formalized file-system crash-consistency models and showed why reasoning about possible post-crash states requires an explicit model [2]. This body of work establishes that recovery claims must name their failure model. It does not, by itself, answer the narrower applied question of how commonly implementable game-save procedures compare under controlled termination of the application process.

This paper evaluates three complete save procedures on one Windows/NTFS system using identical synthetic game-state payloads. It asks: (RQ1) at which save stages does recovery yield the prior complete state, the intended new state, or no valid state? (RQ2) how do the procedures compare in normal save latency and storage footprint? (RQ3) how does recovery routine latency change with payload size? The study contributes a reproducible protocol and harness, a checkpoint-by-checkpoint comparison, and a bounded empirical reference point for the cost and observed process-stop recovery behavior of three save implementations. The source and data are publicly available at public study repository.

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The observed correctness patterns are expected consequences of the procedures and are not claimed as a new algorithm or general durability guarantee.

2. Related Work

Crash-consistency research examines whether application state remains valid when execution stops during storage updates. Pillai et al. studied application update protocols and their dependence on file-system persistence properties [1]. Bornholt et al. developed formal models and validation tools for file-system crash behavior [2]. Their paper analyzes the same temporary-file-then-rename pattern used in replacement-style updates and shows that, under system crashes on some file systems, a rename may persist before the new data, leaving an empty or partial file. Thus, temporary-file replacement alone is not a general power-loss durability guarantee. Min et al. evaluated a file-system interface backed by transactional flash storage [3], and Hu et al. discussed transactions spanning storage abstractions [4]. These works establish that comparing write protocols, integrity checks, and performance is already an active systems topic. The present study is substantially narrower: it does not inject kernel or device failures, infer persistent-media ordering, or model file-system crash states. It measures behavior observable after terminating a writer process at controlled application checkpoints for three game-like local-file procedures on one Windows system.

Game-development documentation provides practical context beyond the systems literature. PlayFab recommends double-buffering local save files as a reliability measure and separately provides cloud conflict handling and rollback [5,6]. Microsoft documents the semantics of the `File.Replace` API, including platform support constraints [7]. Microsoft GDK documentation describes temporary-file replacement as a safe-write pattern [8]. Whittaker's 2009 master's thesis evaluates game-data persistence approaches and reports XML/binary save-load times and file sizes [9]; it does not study controlled interruption recovery. Current Godot documentation demonstrates JSON-based save/load and discusses binary serialization trade-offs, but does not compare interrupted-update recovery [10]. Viotti and Kinderkhedia benchmark JSON-compatible binary serializers using real JSON documents [11], but do not study interrupted game-state updates.

The closest peer-reviewed game-specific systems work is Vaz Salles et al.'s evaluation of checkpoint recovery for massively multiplayer online games [12]. It compares main-memory checkpoint and recovery approaches, including copy-on-update and double-backup disk organization, using simulation and a prototype game server; its recovery model includes checkpointing and log replay after a server crash. Cao et al. subsequently develop checkpoint recovery algorithms for frequently consistent applications, using MMOs as a motivating workload [13]. Zhang's thesis proposes an MMO persistence architecture for durability and fault tolerance, integrates it into the Mammoth research framework, and evaluates the performance of actions implemented with persistence support [14]. Zhang and Kemme also study transaction consistency models for MMOGs, discuss durability and fault tolerance, and measure performance costs [15]. These works establish substantial prior research on game-state persistence and recovery, but concern server-side transaction/state management rather than the local save-file update procedures and process-interruption checkpoints measured here. Knemeyer et al. study relational-database-backed persistence for distributed real-time applications, including MMOGs [16], again at the server-side world/character level. Jeon reports a 2025 arXiv preprint comparing checkpoint installation modes, checksum-based recovery, process-crash injection, and write costs on macOS/APFS [17]. It is the closest methodological comparator located: it also terminates a writer process at checkpoint phases and measures integrity outcomes and performance, though its domain, platform, and update procedures differ. Mustonen's Unity mobile-game thesis describes a save/load implementation using protobuf-net [18], but does not evaluate interruption recovery.

The current study's game-like payload and Windows/.NET setting are contextual replication choices; they do not by themselves establish a strong new method or finding. This is a bounded implementation comparison of local-file behavior under controlled application-process termination. It does not claim that game-save reliability or game-state recovery has been overlooked, nor that the tested procedures are novel. The two-slot method is a practical baseline. The targeted search did not establish absence of prior work, so we make no priority claim; the present distinction is limited to checkpoint-level outcomes and costs for these specific implementations.

The search log records the query families and sources screened. It remains targeted rather than systematic: broad scholarly-database searches, forward/backward citation screening, and independent novelty assessment are still needed before submission. The manuscript therefore makes no priority claim.

3. Materials and Methods

3.1. Save procedures

All strategies began with the same compact UTF-8 JSON payload generated with identical .NET serializer settings. The synthetic state contained player progress, inventory, quest state, and discovered locations. Three procedures were implemented:

1. **Direct overwrite**: truncate the active file and write the new serialized payload in 4,096-byte chunks.

2. **Temporary-file replacement**: create a same-directory temporary file, write and close the new payload, then replace the active file using .NET `File.Replace`.

3. **Two-slot generation**: preserve the prior slot and write the next generation to the alternate slot. Each slot stores a generation number, payload, and SHA-256 checksum. Recovery validates each slot and selects the highest-generation valid slot.

The implementation records its checkpoints and API operations. Replacement behavior is evaluated only on this system; no portable atomicity guarantee is assumed.

3.2. Workload and failure injection

Workloads contained 25, 250, or 1,000 items/locations; the quest count was the greater of one and the integer quotient of item count divided by five. Fixed seeds 7 and 19 generated the old and new states. Within a size, all three strategies used the same expected old/new payloads and hashes. The observed payload sizes were 3,025, 29,510, and 117,963 bytes.

The writer ran as a child process and paused at selected checkpoints before destructive operations, after truncation or slot/temp creation, after each write chunk, after close, and around replacement where applicable. The controller terminated the child at the acknowledged checkpoint. Each strategy/size/checkpoint group was repeated 20 times. After termination, the recovery routine classified the result as old-valid, new-valid, no-valid-save, unexpected-state, or recovery-error. A valid result had to match the expected payload hash and state invariants.

At chunk boundaries the writer called `FileStream.Flush()`, then paused for the controller. This flush moves buffered stream data through the file API, but the study did not request a durable-media flush. The termination mechanism stopped the application process only. The study did not test operating-system crashes, abrupt power loss, storage-controller caches, or physical media failures.

3.3. Measurements and analysis

The fault matrix contained 3 strategies x 3 payload sizes x the applicable predeclared checkpoints x 20 repetitions (3,500 recorded observations overall). Recovery latency measured only the recovery routine and excluded launching the recovery process. Storage bytes recorded the files present after the interruption/recovery path. Normal-save latency used pre-serialized payloads: 10 warm-up saves followed by 100 measurements per strategy and payload size. Timing included file writes, stream flushes, replacement or checksum/envelope work, as applicable; payload generation and JSON serialization were excluded. Strategy order was shuffled with a fixed seed per size.

We report medians and interquartile ranges (IQR) for save latency because the distributions included outliers. Paired median differences compare temporary-file replacement with direct overwrite using 10,000 bootstrap resamples of matched timing observations and a fixed resampling seed. These intervals describe resampling variability within this run only; they are not population intervals over devices, players, or systems. Fault-trial counts are repeated observations at selected checkpoints, not estimates of real-world crash probabilities.

3.4. Unity runtime replication

A separate replication used Unity 6000.3.11f1 and its standalone Windows Player, built-in `JsonUtility` serializer, and `Application.persistentDataPath`. It implemented the same three procedures and workload sizes, with the same deterministic seeds and 4,096-byte chunks; checkpoints followed stream flushes and the controller terminated the Player process after acknowledgment. Unity payloads were 3,027, 29,523, and 118,027 bytes. This matrix contained 147 size/strategy/checkpoint groups with 20 repetitions each, and a separate 900-row timing run with 10 warm-ups and 100 measurements per strategy and size. Recovery used exact Unity-generated payload hashes and state invariants. Four initial 25-item/direct trial directories collided with excluded pilot smoke data because of a controller path bug. The four rows remain visible as controller errors; each was replaced by a new trial 21, yielding 20 valid main trials in each group. Unity files were under C: (WD PC SN560), whereas the .NET main run used D: (Intel SSDPEKNU512GZ); results are analyzed separately and not pooled.

4. Experimental Setup

The .NET main experiment used a standalone C# console harness. It ran in Release configuration on Windows 11 IoT Enterprise LTSC, build 26100 (64-bit), using .NET runtime 9.0.6 and SDK 9.0.301. The processor was an AMD Ryzen 9 7845HX (12 cores/24 logical processors). Save and result files resided on D:, a local NTFS volume on an Intel SSDPEKNU512GZ NVMe drive (nominal 512 GB). The separate Unity runtime replication used Unity 6000.3.11f1 and wrote saves to C:, a local NTFS volume on a WD PC SN560 NVMe drive (nominal 1 TB). Both runs used the same workstation; because their save volumes differed, their timing values are not direct runtime comparisons. The power plan observed after the runs was Balanced; AC/battery state during data collection was not recorded. The .NET primary data

Table 1: *Normal save latency (median, IQR) and post-save directory footprint. Latency values are milliseconds; footprint is bytes.*

Items	Direct overwrite	Temp + replace	Two-slot + checksum	Payload (B)
25	0.120 (0.112–0.132); 3,025 B	1.639 (1.565–1.742); 3,025 B	0.632 (0.161–1.362); 10,810 B	3,025
250	0.156 (0.146–0.232); 29,510 B	5.391 (4.999–5.923); 29,510 B	0.251 (0.238–0.279); 103,380 B	29,510
1,000	0.236 (0.227–0.353); 117,963 B	5.621 (5.186–5.912); 117,963 B	0.536 (0.521–0.582); 412,286 B	117,963

comprise 3,500 fault trials and 900 timing rows; the Unity replication comprises 2,940 valid fault trials, four preserved controller-error rows, and 900 timing rows.

The .NET metadata file labels the protocol as `protocol-v1-draft`, due to a stale metadata constant. Run parameters match protocol version 1, and the discrepancy is documented in the run context; original metadata was not edited. The .NET audit verified expected row/group counts, no duplicates or incomplete groups, and no inconsistent payload hashes or sizes. The Unity audit separately verified 147 complete groups of 20 valid main trials and 900 timing rows; the four pilot-collision records remain in the raw CSV and are excluded from main-trial counts. These audits are data-integrity checks, not independent validation of the experimental design.

5. Results and Discussion

5.1. Recovery outcomes

Across 3,500 interruption observations, recovery returned the old valid state 2,440 times, the new valid state 300 times, and no valid save 760 times. There were no unexpected states or recovery errors. These counts aggregate different checkpoints and strategies and must not be interpreted as crash likelihoods.

For direct overwrite, all repetitions before truncation recovered the old state. Following truncation, the old file was no longer valid; partial writes remained invalid until the complete new payload was present. For temporary-file replacement, the original save remained valid while the separate temporary file was written; recovery selected the new state after replacement. For two-slot saves, the old generation remained available while the alternate slot was incomplete; after a complete checksummed envelope was available, recovery selected the new generation. All 20 repetitions within each tested checkpoint group had the same outcome. This agreement reflects deterministic, controlled checkpoints on one system, not exhaustive fault coverage.

5.2. Normal save latency and storage

Table 1 shows that direct overwrite had the lowest median latency in this run at each payload size. Temporary-file replacement showed a median of 1.639–5.621 ms, while direct overwrite ranged from 0.120–0.236 ms. The two-slot method ranged from 0.251–0.632 ms. The relative ordering and timings are specific to the implementation, runtime, Windows/NTFS stack, and device; no claim is made that the same ranking holds elsewhere.

The paired median latency difference (replacement minus direct) was 1.507 ms (95% paired bootstrap interval 1.474–1.538) for 25 items, 5.030 ms (4.846–5.197) for 250, and 5.174 ms (4.912–5.363) for 1,000. Intervals quantify uncertainty under resampling of these observed pairs only. The two-slot normal-save directory occupied 10,810, 103,380, and 412,286 bytes at the three sizes, respectively, approximately 3.50–3.57 times the base payload. Direct overwrite and replacement occupied approximately one payload equivalent in their normal-save directories. Figure 1 shows the timing distributions and their outliers; Figure 2 compares directory footprint with base payload size. The two-slot footprint is the expected storage cost of retaining two encoded slots and metadata, measured here rather than assumed.

5.3. Recovery routine latency

Median recovery routine time rose with workload size for every strategy. For 25 items, medians were 0.164 ms (direct), 0.150 ms (replacement), and 0.307 ms (two-slot). For 250 items, they were 0.300, 0.288, and 0.763 ms; for 1,000 items, 0.643, 0.781, and 1.511 ms, respectively. These are descriptive measurements across the injected checkpoints. Since checkpoint state and file contents differ, they are not clean isolated estimates of an algorithmic cost, and process startup is excluded.

5.4. Unity replication

The Unity Player produced the same checkpoint-level recovery classification in all 147 groups: across 2,940 valid main trials, 1,880 recovered the old state, 760 had no valid save, and 300 recovered the new state; no trial produced an unexpected state or recovery error. The 25-item/direct pilot collisions are excluded from these counts, with the original four controller-error records preserved and four replacement trials recorded. The pattern matched the .NET study: direct

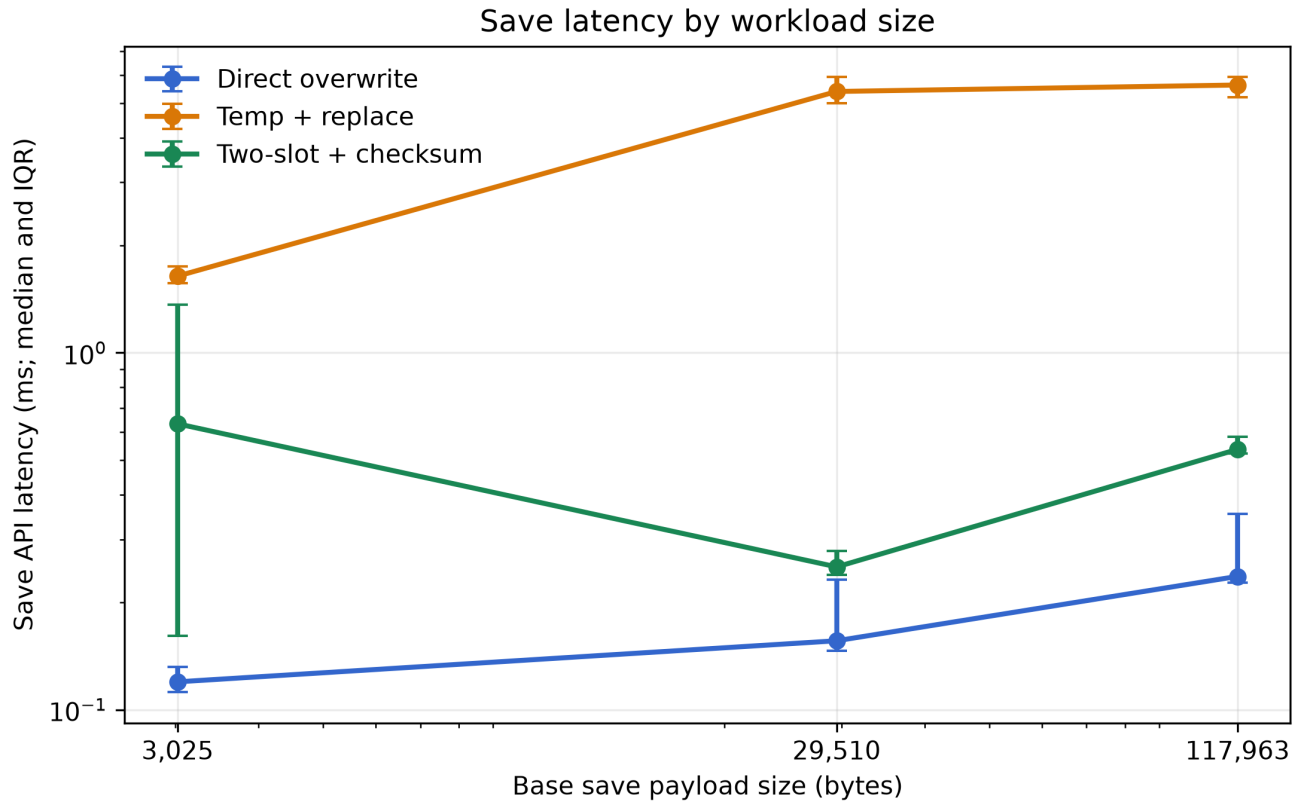


Figure 1: *Normal save latency by payload size and update procedure. Points/distribution summaries reflect 100 measurements per group; the plot shows the observed variability and outliers.*

overwrite exposed an invalid-save interval after truncation, while replacement and two-slot retained a valid older state until commit.

Unity median save latencies were 0.142/0.196/0.316 ms for direct overwrite, 1.678/5.787/5.969 ms for replacement, and 0.238/1.093/2.524 ms for two-slot at 25/250/1,000 items, respectively. The two-slot save directory used 7,177/68,100/271,528 bytes, or about 2.3 times the payload; direct and replacement used approximately one payload. These Unity values are a separate within-runtime/device description: Unity wrote to the C: WD volume, whereas .NET wrote to the D: Intel volume, so absolute timings should not be read as an engine-versus-runtime effect. The agreement in recovery outcomes is a same-workstation replication across different serializers and save volumes, not an independent-platform validation.

5.5. Practical interpretation and validity

The results illustrate a familiar trade-off: direct overwrite is fast and space-efficient during normal saves but exposes a period with no valid file after truncation; the other procedures preserve the prior generation during more of the update, at a time or storage cost. Which trade-off is acceptable depends on a game's state value, expected save frequency, platform, and recovery requirements. The experiment does not measure player harm, perceived reliability, or game-engine integration effort.

Several threats limit interpretation. First, both runs used one workstation and operating system; the two different volumes add a storage-device difference, so timing results are not directly comparable between runtimes. Second, controlled application termination is weaker than power loss or kernel failure; stream flushes did not request durable-media synchronization. Third, a synthetic deterministic JSON workload cannot represent all real game states, compression, encryption, asynchronous save pipelines, cloud synchronization, or platform-specific storage APIs. Fourth, each runtime was run once; repeated timings in one run do not replace independent runs or multiple devices. Fifth, results depend on the specific implementations and checkpoint set. Finally, literature searching was targeted rather than systematic, so no exhaustive novelty claim is justified. A stronger follow-up should repeat the protocol across multiple machines and operating systems, add reboot/power-loss testing where safe, and compare additional engine-native save implementations.

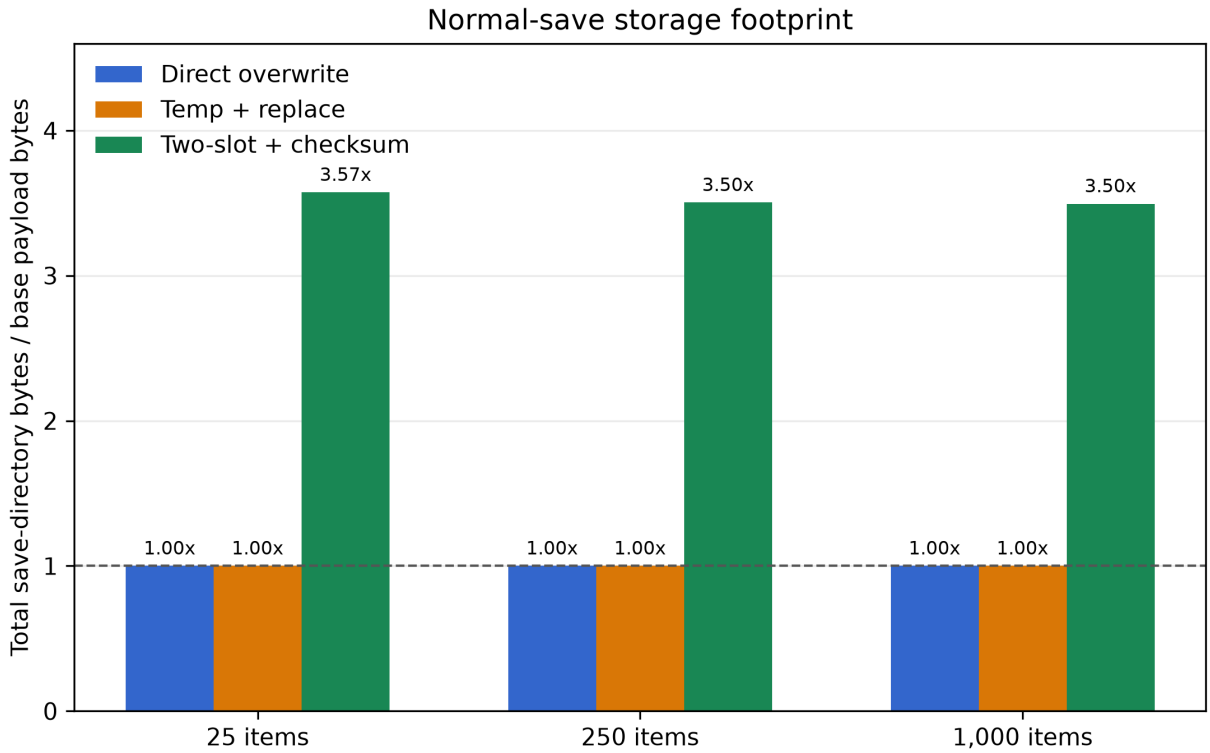


Figure 2: *Normal-save directory footprint relative to the serialized payload size. Two-slot storage retains additional encoded state to preserve a previous generation.*

6. Conclusion and Future Work

This case study compared direct overwrite, temporary-file replacement, and two-slot checksummed saves using controlled process termination and a synthetic game-state workload. The .NET study and separate Unity replication showed the same checkpoint-level recovery pattern: direct overwrite temporarily had no valid save after truncation, while the other strategies preserved a prior generation until a new complete state was available. Direct overwrite had the lowest normal-save latency and storage footprint within each run; the two-slot footprint was about 3.5 times the payload in .NET and 2.3 times in Unity. The timing results are specific to separate save volumes and are not a runtime comparison. These are bounded empirical observations, not platform-independent guarantees.

The immediate next step is independent replication on additional devices and operating systems with durable-flush semantics and machine-level fault injection. Further work should include real engine save paths, larger/compressed payloads, and workload patterns from actual games. The study artifacts are publicly available at the project repository; independent replication and further review are recommended.

Author Declarations

Author Contributions (CRediT): Zain Kassem: Conceptualization, Supervision, Validation, Software. ChatGPT is not an author or CRediT contributor; its assistance is disclosed below.

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Ethics Statement: Not applicable. No human participants, animals, or personal data were involved.

Declaration of Generative AI:

During preparation of this manuscript and study materials, ChatGPT (OpenAI) was used for literature searching, coding assistance, data-analysis support, and manuscript drafting. The author must independently verify all methods, code, sources, results, and wording, revise the manuscript, and accept responsibility for its final content before submission.

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